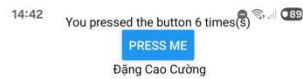


Name:Đặng Cao Cường/2231200087

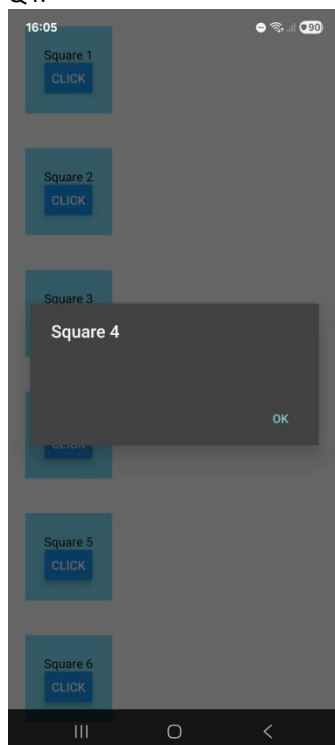
Link to github: <https://github.com/Dang-Cao-Cuong/DangCaoCuongCSW430.git>

Q3:



```
JS App.js U X {} package.json U
Q3Expo1 > JS App.js > default
1 import React,{ useState } from 'react';
2 import { Text,Button, View } from 'react-native';
3
4 export default () =>{
5   const [pressCount, setPressCount] = useState(0);
6   return (
7     <View style={{ alignItems: 'center', marginTop: 20 }}>
8       <Text>You pressed the button {pressCount} times(s)</Text>
9       <Button title="Press me" onPress={() => setPressCount(pressCount + 1)} />
10      <Text> Đặng Cao Cường</Text>
11    </View>
12  );
13 };
14
15
```

Q4:



```
App.tsx M X
Q4ReactNative > App.tsx > App
8 import { NewAppScreen } from '@react-native/new-app-screen';
9 import { StatusBar, StyleSheet, useColorScheme, View ,ScrollView, Button,Alert} from 'react-native';
10 import {
11   SafeAreaProvider,
12   useSafeAreaInsets,
13 } from 'react-native-safe-area-context';
14 import React from 'react';
15 import styles from './style';
16 import data from './Data';
17 import Square from './Square';
18 function App() {
19   const isDarkMode = useColorScheme() === 'dark';
20
21   return (
22     <ScrollView style={styles.container}>
23       {data.map((item,index) => (
24         <Square key={item} text={`Square ${index + 1}`} />
25       ))}
26     </ScrollView>
27   );
28 }
29
30 function AppContent() {
31   const safeAreaInsets = useSafeAreaInsets();
32
33   return (
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\cuong\Downloads\Documents\Q4ReactNative> npx react-native run-android
Starting: Intent { act=android.intent.action.MAIN cat=[android.intent.category.LAUNCHER] cmp=com.q4reactnative.MainActivity }

The collage displays six mobile app mockups, each with a unique design and color palette:

- Employee Information Entry:** Features a dark blue header, a light gray background, and a dark blue success message box. It includes input fields for name, age, and occupation, and a 'CALCULATE' button.
- Calculator:** A simple calculator with a light blue header, a white background, and a blue 'CALCULATE' button.
- Sum First & Last Digit:** A minimalist app with a light gray background, a blue 'CALCULATE' button, and a white input field for the number 254.
- Find Minimum Number:** A light gray app with a blue 'FIND MINIMUM' button and a white input field for the number 238.
- Hailstone Sequence:** A light gray app with a blue 'GENERATE SEQUENCE' button and a white input field for the number 50.
- Success:** A dark gray app with a dark blue header, a light gray background, and a dark blue 'OK' button. It displays a success message and user information.

```

function EmployeeForm(props: {
  fullName: string | undefined;
  onFullNameChange: ((text: string) => void) | undefined;
  age: string | undefined;
  onAgeChange: ((text: string) => void) | undefined;
  occupation: string | undefined;
  onOccupationChange: ((text: string) => void) | undefined;
  onUpdate: ((event: GestureResponderEvent) => void) | undefined;
}) {
  return [
    <View style={styles.formContainer}>
      <Text style={styles.title}>Employee Information Entry</Text>

      <TextInput
        value={props.fullName}
        onChangeText={props.onFullNameChange}
        placeholder="Full Name"
        placeholderTextColor="#000"
        style={styles.input}
      />

      <TextInput
        value={props.age}
        onChangeText={props.onAgeChange}
        placeholder="Age"
        keyboardType="numeric"
        placeholderTextColor="#000"
        style={styles.input}
      />

      <View style={styles.buttonContainer}>
        <Button
          title="Update"
          onPress={props.onUpdate}
          color="#007AFF"
        />
      </View>
    </View>

    function SumFirstDigitAndLastDigit() {
      const [number, setNumber] = useState('');
      const [result, setResult] = useState('');

      const calculateSum = () => {
        if (number === '') {
          setResult('Please enter a number!');
          return;
        }

        const numStr = number.replace('-', '');
        const firstDigit = parseInt(numStr[0]);
        const lastDigit = parseInt(numStr.length - 1);
        const sum = firstDigit + lastDigit;

        setResult('First Digit: ' + firstDigit + '\nLast Digit: ' + lastDigit + '\nSum: ' + sum);
      };

      return [
        <View style={styles.formContainer}>
          <Text style={styles.title}>Sum First & Last Digit</Text>

          <TextInput
            value={number}
            onChangeText={setNumber}
            placeholder="Enter a number"
            placeholderTextColor="#000"
            keyboardType="numeric"
            style={styles.input}
          />

          <View style={styles.buttonContainer}>
            <Button
              title="Calculate"
              onPress={calculateSum}
              color="#007AFF"
            />
          </View>

          function FindMinimum() {
            const [num1, setNum1] = useState('');
            const [num2, setNum2] = useState('');
            const [num3, setNum3] = useState('');
            const [result, setResult] = useState('');

            const findMinimum = () => {
              if (num1 === '' || num2 === '' || num3 === '') {
                setResult('Please enter all three numbers!');
                return;
              }

              const n1 = parseFloat(num1);
              const n2 = parseFloat(num2);
              const n3 = parseFloat(num3);

              let min = n1;
              if (n2 < min) {
                min = n2;
              }
              if (n3 < min) {
                min = n3;
              }
            };
          }
        </View>
      ];
    }
  ];
}

```

```

126     setResult('Numbers: ' + n1 + ', ' + n2 + ', ' + n3 + '\nMinimum: ' + min);
127   });
128
129   return (
130     <View style={styles.formContainer}>
131       <Text style={styles.title}>Find Minimum Number</Text>
132
133       <TextInput
134         value={num1}
135         onChangeText={setNum1}
136         placeholder="First Number"
137         placeholderTextColor="#000"
138         keyboardType="numeric"
139         style={styles.input}
140       />
141
142       <TextInput
143         value={num2}
144         onChangeText={setNum2}
145         placeholder="Second Number"
146         placeholderTextColor="#000"
147         keyboardType="numeric"
148         style={styles.input}
149       />

```

```

1     <TextInput
2       value={num3}
3       onChangeText={setNum3}
4       placeholder="Third Number"
5       placeholderTextColor="#000"
6       keyboardType="numeric"
7       style={styles.input}
8     />
9
10    <View style={styles.buttonContainer}>
11      <Button
12        title="Find Minimum"
13        onPress={findMinimum}
14        color="#007AFF"
15      />
16    </View>
17
18    <Text>{result}</Text>
19  </View>
20  );
21  }

```

```

function HailstoneSequence() {
  const [number, setNumber] = useState('');
  const [sequence, setSequence] = useState('');

  const generateHailstone = () => {
    if (number === '') {
      setSequence('Please enter a number!');
      return;
    }

    const n = parseInt(number);
    if (n <= 0) {
      setSequence('Please enter a positive number!');
      return;
    }

    let current = n;
    let steps = [current];

    while (current !== 1) {
      if (current % 2 === 0) {
        current = current / 2;
      } else {
        current = current * 3 + 1;
      }
      steps.push(current);
    }
  }
}

```

```

201   const result = 'Starting Number: ' + n + '\nTotal Steps: ' + steps.length
202   + '\n\nSequence:\n' + steps.join(' - ');
203   setSequence(result);
204 };
205
206   return (
207     <View style={styles.formContainer}>
208       <Text style={styles.title}>Hailstone Sequence</Text>
209
210       <TextInput
211         value={number}
212         onChangeText={setNumber}
213         placeholder="Enter a positive number (n > 0)"
214         placeholderTextColor="#000"
215         keyboardType="numeric"
216         style={styles.input}
217       />
218
219       <View style={styles.buttonContainer}>
220         <Button
221           title="Generate Sequence"
222           onPress={generateHailstone}
223           color="#007AFF"
224         />
225       </View>
226
227       <Text>{sequence}</Text>
228     </View>
229   );

```

```

232 function App() {
233   const [fullName, setFullName] = useState('');
234   const [age, setAge] = useState('');
235   const [occupation, setOccupation] = useState('');
236
237   const handleUpdate = () => {
238     if (fullName === '' || age === '' || occupation === '') {
239       Alert.alert('Error', 'Please fill in all fields!');
240       return;
241     }
242
243     const message = 'Update data successfully!\n\nName: ' + fullName
244     + '\nAge: ' + age + '\nOccupation: ' + occupation;
245     Alert.alert('Success', message);
246   };
247 }

```

```

248   return (
249     <ScrollView style={styles.container}>
250
251       <EmployeeForm
252         fullName={fullName}
253         age={age}
254         occupation={occupation}
255         onFullNameChange={setFullName}
256         onAgeChange={setAge}
257         onOccupationChange={setOccupation}
258         onUpdate={handleUpdate}
259       />
260
261       <SumFirstDigitAndLastDigit />
262       <FindMinimum />
263       <HailstoneSequence />
264     </ScrollView>
265   );
266 }

```

```

267 const styles = StyleSheet.create({ 298
268   container: { 299
269     flex: 1, 300
270     backgroundColor: '#f5f5f5', 301
271     marginBottom: 20, 302
272   }, 303
273   formContainer: { 304
274     padding: 20, 305
275     backgroundColor: '#fff', 306
276     margin: 20, 307
277   }, 308
278   title: { 309
279     fontSize: 20, 310
280     fontWeight: 'bold', 311
281     marginBottom: 20, 312
282     textAlign: 'center', 313
283     color: '#0c0c0cff', 314
284   }, 315
285   input: { 316
286     padding: 12, 317
287     borderWidth: 1, 318
288     borderColor: '#ccc', 319
289     borderRadius: 8, 320
290     marginBottom: 15, 321
291     backgroundColor: '#fafafa', 322
292     fontSize: 16, 323
293   }, 324
294   buttonContainer: { 325
295     marginTop: 30, 326
296   }, 327
297   resultContainer: {
298     marginTop: 20,
299     padding: 15,
300     backgroundColor: '#e8f5e9',
301     borderRadius: 8,
302     borderWidth: 1,
303     borderColor: '#4caf50',
304   },
305   resultText: {
306     fontSize: 16,
307     color: '#2e7d32',
308     lineHeight: 24,
309   },
310   infoBox: {
311     backgroundColor: '#fff3cd',
312     padding: 15,
313     borderRadius: 8,
314     marginBottom: 15,
315     borderLeftWidth: 4,
316     borderLeftColor: '#ffc107',
317   },
318   infoText: {
319     fontSize: 14,
320     color: '#856404',
321     marginBottom: 3,
322   },
323 });
324
325 export default App;

```